Small Engines Overhaul · Unit 5 · Level 2 (modified)

Carburetion & Fuel



TODAY'S PLAN

- 💧 Purpose of the carburetor
- ⏳ The venturi
- 🔧 Carburetor parts and types
- ⛽ Fuel, oil and ethanol

Review · 5 minutes

Quick Review



Name 3 parts that seal the cylinder.

Where does the compression gauge go?

What does PSI stand for?

Words to Know

Words to Know



Carburetor

The part that mixes air and fuel in the right amounts for the engine.



Venturi

A narrow part of the carburetor throat where air speeds up and pressure drops, pulling fuel in.



Choke

A plate that limits air for a richer mix when starting a cold engine.



Throttle

A plate that controls how much air-fuel mix enters, changing engine speed.

Words to Know

More Words to Know



🛟 **Float**: A part that floats on fuel in the bowl and shuts off fuel when the bowl is full.

🌽 **Ethanol**: Alcohol mixed into gasoline (E10). It attracts water and can gum up small engines.

🧪 **Fuel stabilizer**: An additive that keeps stored gas from going bad.

⚖️ **Air-fuel ratio**: The mix of air to fuel, about 14.7 parts air to 1 part fuel by weight.

Learn

Purpose of the Carburetor



Mixes **air and fuel** in the right ratio (about 14.7:1 by weight).

Turns liquid fuel into a fine mist that burns.

Controls speed with the **throttle**.

Learn

The Venturi Effect



The carburetor throat narrows at the **venturi**.

Air speeds up there and its **pressure drops**.

Higher air pressure on the fuel bowl pushes fuel up the jet into the airstream.

🃏 Check 1 of 2



Air in the venturi…

Hold up 1, 2, or 3:

1️⃣ Slows down and pressure rises

2️⃣ Speeds up and its pressure drops

3️⃣ Stops

✅ Answer · Card 1 of 2



Speeds up and its pressure drops

Fast air, low pressure.

Learn

Carburetor Parts



Choke
less air for cold start



Throttle
controls speed



Float + needle
keeps fuel level



Jet
meters fuel

Compare

Types of Carburetors



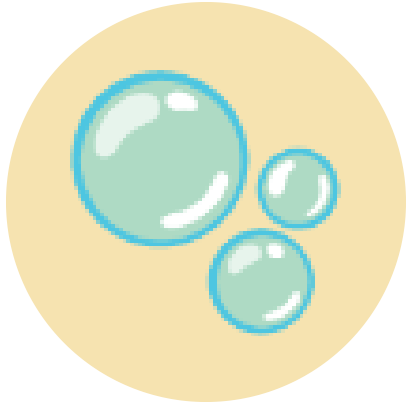
Float type

Fuel bowl with a float
Most lawn mowers

Diaphragm type

Rubber diaphragm pumps fuel
Chainsaws and trimmers, any angle

🃏 Check 2 of 2



A carburetor that works at any angle is…

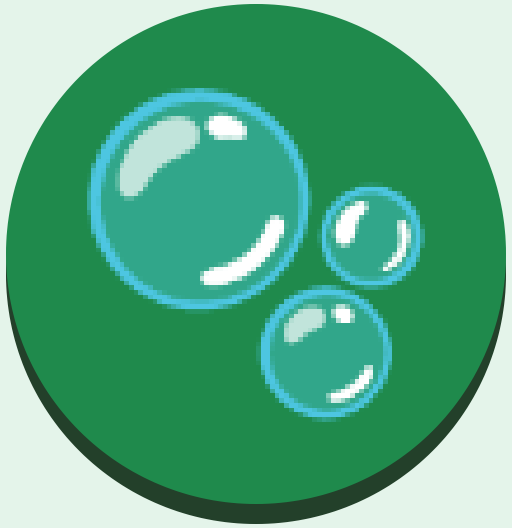
Hold up 1, 2, or 3:

1️⃣ Float type

2️⃣ Gravity only

3️⃣ Diaphragm type

✅ Answer · Card 2 of 2



Diaphragm type

Chainsaws tip over.

Learn

Fuel and Oil Care



Ethanol (E10) attracts water and gums carburetors; use fresh fuel or stabilizer.

2-strokes need the **right oil mix** (like 50:1).

Store with stabilizer or run the carburetor dry.

Wrap-Up

Recap



Carburetor
air + fuel



Venturi
fast air, low pressure



Choke
cold start



Ethanol
can gum up

Wrap-Up

Exit Ticket



What does the carburetor do?

Explain the venturi in one sentence.

Name one fuel care tip.